NDRA-PII Developer Report

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Project: Neuro-Semantic Distributed Risk Analysis for Personally Identifiable Information (NDRA-PII)
Date: December 19, 2025
Current Phase: Phase 3 Complete (Ready for Phase 4 Fusion)

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1. Executive Summary

This report documents the architectural implementation of the NDRA-PII system. As of today, the system has successfully evolved from a conceptual design into a functional **Multi-Model Intelligence Platform**. We have established a strictly governed, agentic pipeline capable of ingesting diverse enterprise data, ensuring cryptographic integrity, and performing high-precision PII detection with granular location mapping.

2. System Architecture Implemented

The system follows a **Deterministic Multi-Agent Architecture** where each agent has a single, isolated responsibility.

2.1 Core Infrastructure (`core/`, `schemas/`)

- **Strict Typing**: Implemented Pydantic models (`DocumentMetadata`, `SemanticChunk`, `DetectedPII`) to enforce data consistency across agents.
- **Immutable Audit**: The `AuditAgent` (`agents/audit.py`) is active, recording every system event (Upload, Extraction, Detection) with **SHA-256 cryptographic chaining**. This ensures that the reasoning history is tamper-evident.
- **Service Layer**: A `FastAPI` application (`main.py`) serves as the entry point, supporting `multipart/form-data` uploads and real-time analysis.

2.2 Perception Layer: `ExtractorAgent` (`agents/extractor.py`)

We moved beyond basic PDF parsing to a robust **Multi-Model Ingestion Engine**.

1. **Supported Formats**:
    1. **Documents**: PDF, DOCX, PPTX.

2. **Data**: XLSX, CSV, JSON, XML, YAML.
3. **Communication**: EML, MSG.
4. **Web**: HTML.
5. **Archives**: ZIP, TAR.

2. Key Features:

1. **Integrity**: Every file is SHA-256 hashed upon entry.
2. **Semantic Chunking**: Text is split into sliding windows (e.g., 500 tokens) respecting sentence boundaries to preserve context for NLP.
3. **Quarantine**: Files with unsupported MIME types are safely isolated to a `/quarantine` directory.

2.3 Cognition Layer: `ClassifierAgent` (`agents/classifier.py`)

This is the "Brain" of the PII detection capability.

1. **Engine**: Integrated **Microsoft Presidio** using the **Spacy `en_core_web_lg`** (Large) model for superior Named Entity Recognition (NER).

2. **Custom Recognizers**: Added specific regex patterns for Indian Context (**Aadhaar**, **PAN**) alongside standard US/Global entities.

3. **Granular Mapping** (New Feature): - The system now maps every detected PII entity to its **Exact Location**.

4. **Output Example**: `Page 32 [Offset 450:461]`. - **Context**: Captures a "snippet" of text around the PII for explainability.

2.4 Interfaces

- **REST API**: endpoints (`/analyze/upload`) for integration.
- **CLI Tool**: `ndrapiicli.py` for rapid, local testing without server overhead.

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4. Verification & Testing status

Test Suite	Scope	Status	Result
`test_phase2.py`	PDF Ingestion & Hashing	**PASS**	Validated SHA-256 & Chunking
`test_real_pdf.py`	Real Document Extraction	**PASS**	Processed `Testing_Set.pdf`
`test_phase3_real.py`	PII Detection (PDF)	**PASS**	Found SSNs, Credit Cards, Emails

`test_multi_model.py`	Excel/JSON Ingestion	**PASS**	Processed `Testing_Set.xlsx`
`test_excel_pii.py`	Detailed Excel Mapping	**PASS**	**971 PII Entities** mapped with offsets

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4. Next Steps (Development Plan)

We are now transitioning from **"Observation"** (Seeing PII) to **"Governance"** (Acting on PII).

Phase 4: PII Fusion (Immediate)

- **Goal**: Clean duplicate detections (e.g., overlapping spans).
- **Architecture**: A logic layer within `ClassifierAgent` to merge Entities based on score and span.

Phase 5: Governance (NSRL)

- **Goal**: Make decisions.
- **Mechanism**: **Policy Agent** (`agents/policy.py`).
- **Input**: `ClassifiedChunk`.
- **Logic**: Neuro-Symbolic Rule Layer (YAML Rules).
  - *Example*: `IF entity="Aadhaar" THEN action="Redact"`.

Phase 6: Redaction

- **Goal**: Enforcement.
- **Mechanism**: `Presidio Anonymizer` to mask text at the identified offsets.

Report Status: Complete.
Signed: NDRA-PII Agent.